MACHINE PSYCHOLOGY RESEARCH PROGRAMME

# Identity-Reactive Conversational Behaviors in Large Language Models: An Exploratory Multi-Model Study of Name-Associated Cultural and Conversational Responses

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**KEY RESULTS**

**135** response observations in 27 sessions

**10** candidate phenomena catalogued

**0** coded overt-harm instances

**7/27** Cultural Boxing Food, all Reza Moradi

## AT A GLANCE

- Prompts identical except for the user's name still produced responses differing in cultural inference, empathy, language use and topic engagement across three frontier models.
- Zero coded instances of the predefined overt-harm category across all 135 responses.
- Ten candidate identity-reactive phenomena catalogued, all framed as exploratory constructs that require replication.

## INTRODUCTION

Fairness evaluation of large language models often emphasizes explicit harms. Yet a user's name operates below that threshold: it makes no explicit demographic statement, but can carry cultural associations a model may react to.

This study asks whether responses change when matched prompts carry different name-associated identity cues (RQ1), how patterns vary across models, names and domains (RQ2), and what follows for fairness evaluation (RQ3).

## METHODS

- **Design.** 3 models × 3 names × 5 domains × 3 registers = 135 observations in 27 fresh sessions.
- **Models.** ChatGPT-5.4, Claude 4.6 Sonnet and Gemini 3.1 Pro, via public web interfaces, April 2026.
- **Names.** Jake Thompson, Tyrone Williams and Reza Moradi: identity-associated cues, not validated measures of ethnicity.
- **Domains.** Food, Childcare, Lost Wallet, Refugee, Airport. **Registers:** formal, moderate, informal.
- **Analysis.** Structured human coding; session-matched Friedman tests, Benjamini-Hochberg adjusted.

## RESULTS IN DETAIL

- **Cultural Boxing (CB).** Every coded case sat in the Reza Moradi conditions; Gemini framed 3/3 variants as Persian/Iranian cuisine, never inferred for the other two names.
- **Linguistic Mirroring (LM).** Claude 6/45, Gemini 13/45. Gemini also answered the Tyrone Williams condition in Persian (Cultural Misattribution Error, CME).
- **Friedman tests.** Response length $\chi^2 = 14.00$, $p = .00091$; Proactive Empathetic Shielding $\chi^2 = 14.55$, $p = .00069$; human-coded empathy $\chi^2 = 9.25$, $p = .00980$. All survived Benjamini-Hochberg adjustment ($q < .05$).

| Group | Candidate phenomena |
|---|---|
| **Cultural inference** | Cultural Boxing (CB), Unmarkedness Paradox (UP), Cultural Misattribution Error (CME) |
| **Empathetic differentiation** | Default Empathetic Amplification (DEA), Proactive Empathetic Shield (PES), Inverted Empathy Spectrum (IES) |
| **Linguistic adaptation** | Linguistic Mirroring (LM), Cultural Lexical Sensitivity (CLS), Lexical Surface Sensitivity (LSS) |
| **Topic engagement** | Topic Avoidance (TA), strongest in Gemini Airport responses |

## CONCLUSIONS

- Models can produce systematically differentiated responses to identity-associated cues without producing overtly derogatory content.
- A harm/no-harm check does not exhaust what is worth auditing.
- The candidate taxonomy is a hypothesis-generating framework for replicated testing, not a validated set of constructs and not evidence of stable model personalities.

## LIMITATIONS

- Responses are nested within sessions, and fixed domain order confounds domain with session position.
- One observation per factorial cell: no within-cell replication.
- The three names are identity-associated cues, not independently pretested measures of ethnicity.
- Stimulus design and initial pattern identification were led by a single author; no human baseline was collected.

## WHY IT MATTERS

- Safety auditing can extend beyond explicit toxicity to whether name-based inference alters register, empathy or advice quality.
- All 27 session transcripts, feature matrices and the analysis pipeline are openly archived, making the framework directly reusable.